Cheat sheet

Vectors & RAG

Embeddings, vector databases, retrieval and evaluation

Pipeline

1 Ingest — 2 Parse — 3 Chunk — 4 Embed — 5 Index — 6 Retrieve —
7 Rerank — 8 Generate — 9 Evaluate

Design choices

Chunking	Fixed, recursive, structure-aware, semantic; 300–800 tokens, 10–20% overlap
Embedding model	Quality on your language/domain, dimensions, cost, speed
Index	HNSW (fast, memory-heavy), IVF (clustered), flat (exact, small data)
Distance	Cosine, dot product, Euclidean (normalise vectors for cosine)
Hybrid	BM25 + dense; fuse with reciprocal rank fusion
Reranker	Cross-encoder over top-N candidates
Metadata	Source, page, section, date, tenant, ACL; filter before ranking
Context assembly	Deduplicate, order by relevance, cite sources, respect token budget

Advanced

Query rewriting	Clarify / expand the question
Multi-query / RAG-Fusion	Several queries, merged results
HyDE	Embed a hypothetical answer
Parent-child	Retrieve small, return larger parent
Contextual retrieval	Prepend chunk-specific context before embedding

GraphRAG	Entity graph + community summaries
Agentic RAG	Agent decides searches iteratively

Evaluation

Retrieval	Recall@k, precision@k, MRR, nDCG, hit rate
Generation	Faithfulness, answer relevance, context precision/recall, citation accuracy
Tools	RAGAS, DeepEval, TruLens, Arize Phoenix (open source)

⚠ **Watch out**

Common failures: bad PDF parsing, chunks without context, stale indexes, missing permissions filtering, too many chunks (noise), no abstain path.